

Reflection on Learning and AI Use

Module: IE3090 Network Programming

Assignment: RemoteOps (Part 1)

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DRAFT NOTE (delete before submitting): the facts below come from our conversation, but a reflection is marked on being specific and honest about YOUR experience, and you are examined on it. Rewrite everything in your own words, fill every yellow [bracket], keep the final text between 300 and 500 words, and remove this note.

1. Which AI tools did I use, and at which stages?

I used Claude (Anthropic) throughout Part 1. Early on it helped me connect my CentOS 10 machine to GitHub over SSH and divide the assignment into a three-day plan. During implementation it drafted the C code for the Agent and Controller in segments, which I built, ran and committed one at a time. Near the end I used it to decide which evidence to capture, to generate the architecture diagram, and to structure the Implementation Report. [Add any other tool or stage, or delete this sentence.]

2. What did the AI do well, and where did it go wrong?

It was strongest on structure: the personalised values, the line-based framing with a per-session buffer, and a single reply function that always appends SID:0043. It also gave me test commands, such as sending AUT and then H OPS-3400 a second later, that proved the framing worked. It was weaker about my own setup. It assumed I could download files straight onto the CentOS machine, but my chat and the VM are separate, so I had to copy and paste instead. It also assumed my registration number from my GitHub username, and its line numbers for code screenshots moved after later edits. [Add a bug or wrong suggestion you personally ran into, and how you noticed it.]

3. What did I change, add or reject, and why?

I did not keep code I could not explain. I shortened and reworded comments, renamed agent.c to agent_400.c to meet the naming rule, and tested the results myself, including sha256sum and cmp on PUT and GET and five simultaneous clients. [Describe at least one specific change you made to AI-written code, or one suggestion you rejected, and give your reason.]

4. What did I learn about my own understanding of network programming?

[Write 4 to 6 sentences in your own words. Ideas you can draw on: TCP is a byte stream, so one recv() can hold half a line or several lines, and testing read_line() with AUT then H OPS-3400 showed this; why the first bytes of a PUT can arrive in the same recv() as the command line, and why leftover bytes must be kept; why a rejected PUT must discard the incoming bytes; why the SID has to be a string; what thread-per-client taught you about shared state (the log mutex) and cleanup (stopping the monitor thread on every exit path); and what you would do differently next time.]

Overall, AI saved time on boilerplate, but understanding the protocol well enough to explain it and extend it live was my own work. [Edit this closing sentence so it is true for you.]